Volume V · Chapter 23 — Robotics Fundamentals & Simulation (ROS 2, Isaac Sim) · Labs

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-V-Chapter-23-context.md` for the AI interaction log.

Prerequisites: Ch 4; general programming maturity.

Learning outcomes — the student can: explain robot kinematics/control basics; use ROS 2 concepts (nodes, topics, services, actions); run Gazebo/Isaac Sim; command a simulated arm.

Labs (hands-on) — 16 h:

#	Lab	h
23a	Set up ROS 2 + a simulator; run a demo	4
23b	Publish/subscribe: command a simulated arm via topics	4
23c	Bring up a simulated robot arm in Isaac Sim; basic motion	4
23d	Scripted pick task in simulation	4

Datasets/tools: ROS 2, Gazebo, Isaac Sim; GPU workstation. **Assessment:** simulation task deliverable (**60%**); ROS 2 quiz (**20%**); writeup (**20%**). **Key decisions:** simulator choice; **sim vs. hardware**; real-time constraints. **References:** plan §7; §13 → *Robotics & embodied AI*. **Hours:** Theory **10** + Lab **16** = **26**.